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Development of Air Combat Effectiveness Simulation and Analysis Scheme for Beyond Visual Range (BVR) Case

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Abstract. This paper focuses on the simulation of *Beyond Visual Range (BVR) Air Combat* using calculation of Probability of Kill (PK) and Supremacy to analyze the effectivity of the tactics and strategies. The developed simulation program is expected to be used as a means of developing tactics, weapons systems evaluation and to support other air combat system applications. The simulation system can be utilized so that strategy for using weapons and better tactics for maneuver can be determined and formulated in the face of an aerial battle. The simulation illustrates how an aerial battle in a 2D plane took place. The battle occurs between two sides and can also cover case of many against many air battle. Fighters maneuver will be generated based on logic and defined tactic by providing input in the form of initial conditions. Formulation / functions, parameters and variables influencing the fighter effectiveness will be taken into account. These elements are modelled using a dynamic and open scenario approach. The simulation used a method called 'Missile Launch Envelope Model' for computing PK of missile along its way to the target. Simulation results for one versus one and many versus many cases show that the developed system can produce good and relatively realistic prediction of the outcome. Further modification of simulation program include fly out modeling to the missile.

Introduction

To develop fighter aircraft technology aiming at gaining air superiority, the study of the effectiveness of the tactics and strategies of air combat becomes very important. There are many factors that affect the level of effectiveness of an engagement in air combat (combat effectiveness), such as tactical maneuvers, aspects of the relative position and distance to the enemy aircraft, differences in altitude, speed, energy management, weaponry, guidance systems, sensors and radar capability, weather conditions, etc.[1]. These factors are interrelated and influenced by each other, depending on the circumstances when the combat takes place. To quantify the air combat effectiveness in an aerial battle, a probability value is defined and formulated, which is called *Probability of Kill (PK)*. From the target standpoint (defensive side), the ability or chance to withstand the attack, denoted by *Probability of Survive (PS)*, must also be considered. Some factors that affect the *Probability of Survive (PS)* including enemy stealth technology, the ability of early detection, countermeasures, aircraft maneuver, etc[1]. Further, to describe the potential a fighter has to overpower or outperform its enemy, another parameter called *Supremacy* is defined. The *supremacy* level of a fighter over its enemy can be used for planning the engagement maneuver and for selecting the most vulnerable target among several possible enemies. Supremacy can be viewed as a measure to indicate the level of advantage one fighter has over its potential opponents.

Some works focusing on the analysis of air combat effectiveness, both in Within Visual Range (WVR) and Beyond Visual Range (BVR) cases, can be found in some literatures, reflecting the rapid technology development in this area. PK formulation and variables that affect have been extensively elaborated in [1] which focuses more on WVR case, some of which become the basis for the formulation developed in this research. The description and definition of aircraft *supremacy* are discussed in [2], which also proposed some formulations for quantifying the parameters. While in references [3] and [4] some maneuver strategies for BVR air combat that defined by taking into

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account supremacy aspects are discussed and examined. In weaponry side, the detailed data about some missile commonly used in BVR air combat can be found in reference [5]. Further, to analyze and model the performance of missile weaponry, Kaplan [6] has described an approach for determining the missile flyout, an area which can be reached by a missile corresponding to its characteristics and launch condition.

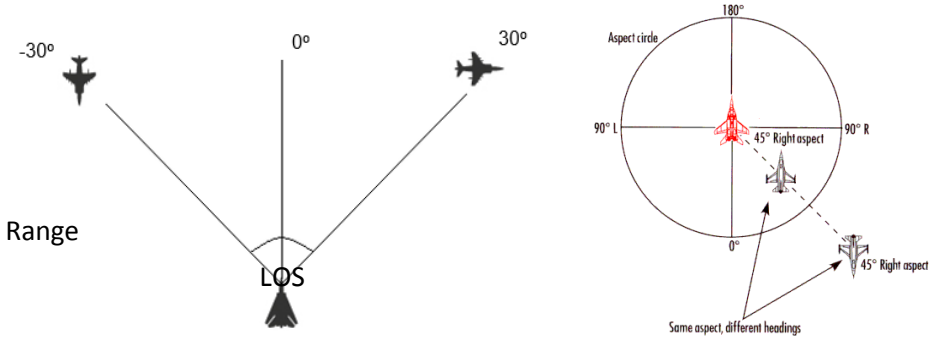
In this paper, the development of an air combat simulation system, which can be used as a tool for evaluating the effectiveness of maneuver, tactics, and weaponry in a combat situation, is presented. The development involves the modeling of fighter maneuver, definition and formulation of air combat effectiveness parameters/measures, usually known as Probability of Kill (PK), and simulation outcome logic. Calculation of PK and supremacy mentioned above will be assessed by empirical methods, which are implemented in MATLAB® [7]. The numerical simulation platform, which is also built on MATLAB/SIMULINK environment, later on will also be integrated with flight simulation software.

The paper is presented in the following arrangement. The second section discusses the description and definition related to air combat modeling parameters, followed by the third section that elaborates air combat effectiveness parameters, i.e. PK and supremacy. The fourth section presents some features of the developed simulation system. The next section will be focused on the implementation of the simulation scheme for analyzing some combat scenarios. The sixth section concludes the paper with some remarks.

Air Combat Modeling

Some aspects and parameters that are used for determining the combat effectiveness and become key points in formulating the PK and Supremacy must be clearly defined.

Positional Geometry. Positional geometry is defined as the relative position between the fighter and the bandit in the battlefield. Positional geometry parameters are used to determine the level of PK and supremacy of a fighter with respect to its opponent. The geometrical parameters mainly used for determining PK and supremacy are Range, Line of Sights (LOS) Angle, and Aspect Angle.



(a) (b)

Figure 1. Positional Geometry parameters (a) Range and LOS, (b) Aspect angle

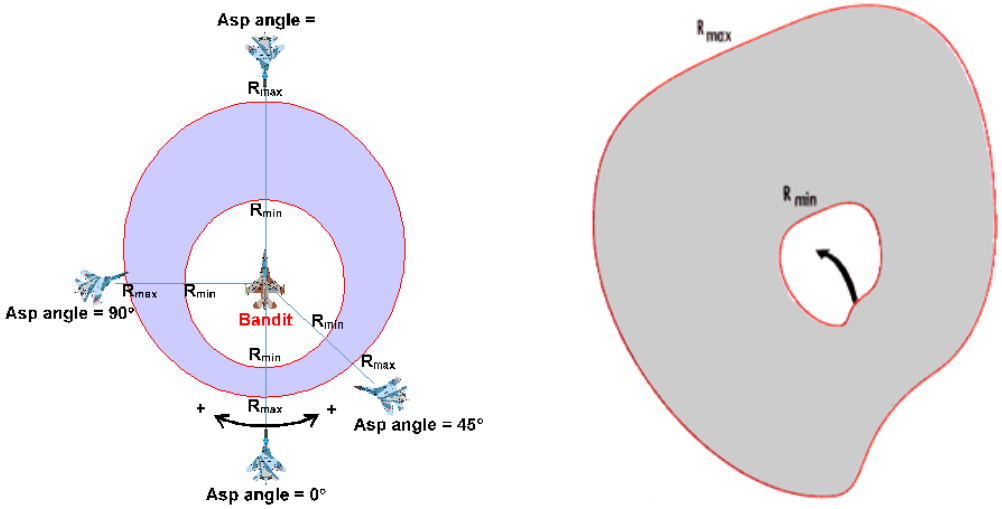
Weaponry. The type and capability of the weapon carried by a fighter will significantly define the combat capability of the fighter. Some missiles such as Air-to-air missile (AAM) becomes standard weapon for some fighters. This type of missile can be categorized based on its range of operation, such as short range, medium range, and long range missiles. Detailed information on these type of missiles along with its variation of guidance can be found in reference [1].

Missile Envelope. The capability of a missile in a combat also determined by its maximum and minimum range (Rmax & Rmin). The missile range can be illustrated in a diagram called missile envelope. Missile envelope is the area around the bandit (opponent) where the use of missiles become effective (missiles launched inside this area are most likely to hit the target). The missile envelope is affected by some factors, such as types of missiles, the aspect angle when the missile is fired, target aircraft (bandit) G-load, altitude, and fighter speed. In the head on case, the missile can be launched from a longer distance from the opponent, compared to that in tail chase situation. This relation can be formulated as follow.

$$R_{max} = R_{max-TC} + \frac{Aspect\ angle}{180}.(R_{max-HOC} - R_{max-TC}) \quad (1)$$

$$R_{min} = R_{min-TC} + \frac{Aspect\ angle}{180}.(R_{min-HOC} - R_{min-TC}) \quad (2)$$

where $R_{max\text{-}HOC}$ and $R_{min\text{-}HOC}$ are missile maximum and minimum attained range, respectively, when fired in front of bandit (Head-On Chase). While $R_{max\text{-}TC}$ and $R_{min\text{-}TC}$ are missile maximum and minimum attained range, respectively, when fired behind the bandit (*Tail Chase*).
The aircraft flight altitude will also affect the range of the missile. Higher altitude will result in a larger missile envelope, due to reduced missile drag. The lower altitude the air combat take places, the smaller the missile envelope becomes.



(a) (b)

Figure 2. Missile Envelope (a) function of Aspect angle, (b) function of g-load manuver [1]

Air Combat Effectiveness

Probability of Kill (PK). PK model/formulation was designed using geometrical approach by considering the change of relative position variables at each time instant due to fighter and bandit maneuvers. These positional changes are characterized by the value of dynamic geometry variables such as distance, aircraft speed, LOS angle and aspect angle when missile fired. Considering the main variables, i.e. range and azimuth angle, the PK can be formulated as

$$PK = P_r \cdot P_a \quad (3)$$

where,

$$P_r = \begin{cases} 0 & , R_b < R_{min} \\ 1 - \frac{R_{Kmin} - R_b}{R_{Kmin} - R_{min}} & , R_{min} \le R_b < R_{Kmin} \\ 1 & , R_{Kmin} \le R_b \le R_{Kmax} \\ 1 - \frac{R_b - R_{Kmax}}{R_{max} - R_{Kmax}} & , R_{Kmax} < R_b \le R_{max} \\ 0 & , R_b > R_{max} \end{cases}$$

$$P_a = \begin{cases} 1 - \frac{\lambda_f}{\varphi_W} & , \lambda_f \le \varphi_W \\ 0 & , \lambda_f > \varphi_W \end{cases}$$

P_r = *Probability of kill by range*
P_a = *Probability of kill by angle*
R_b = *Bandit range*
R_{min} = *Missile minimum distance*
R_{Kmin} = *Missile minimum kill distance*
R_{Kmax} = *Missile maximum kill distance*
R_{max} = *Missile maximum distance*
λ_f = *LOS angle*
φ_W = *Maximum boresight angle*

Supremacy. In a BVR air combat, the effectiveness of the tactics and maneuvers can be measured from the level of supremacy the fighter can gain against the bandit. The concept of supremacy can also be viewed as the level of potential or advantages possessed by a fighter such that it can successfully engage and destroy its opponent (by gaining high PK value). The supremacy is determined by some factors such as relative positioning, weapon capability, detection system capability, energy advantage (potential and kinetic), etc. In addition to assessing the effectiveness of the maneuver or positioning, *supremacy* can also exploited for assisting the pilot to execute the target selection task, i.e. to select the most vulnerable opponent to be targeted in case of multi-target engagement. There are some different formulations of *Supremacy*, such as that described in [1], while in this paper the following formulation is proposed:

$$S = \alpha_s (\beta_d S_d + \beta_\lambda S_\lambda + \beta_q S_q) + \alpha_e S_e$$
$$\alpha_s + \alpha_e = 1$$
$$\beta_d + \beta_\lambda + \beta_q = 1 \quad (4)$$

In Eq.4, S is *Total supremacy*, S_d is *Distance supremacy*, S_λ is *Azimuth supremacy* (λ=*LOS angle*), S_q is *Impact angle supremacy*, and S_e is *Energy supremacy*. Further, the coefficients α_s, α_e, β_d, β_λ, β_q denote the weights for *position supremacy, energy supremacy, distance supremacy, azimuth supremacy,* and *impact angle supremacy*, respectively.

Air Combat Simulation System

Air Combat Simulation Environment. Based on the definitions and features that forming the components of the air combat model, along with the formulation of combat effectiveness parameters (PK and supremacy), a simulation scheme is developed for simulating and analyzing combat situation. The system is developed using MATLAB programming language [7], whose functions and logic are represented in Figure 3.

The simulation can be initialized by setting the input via the developed software graphical user interface (GUI), as depicted in Figure 4. The main GUI allows users to input the scenario of the air combat along with combat parameters settings, such as weapon firing setting, maneuver setting, radar range, weapon range, etc.

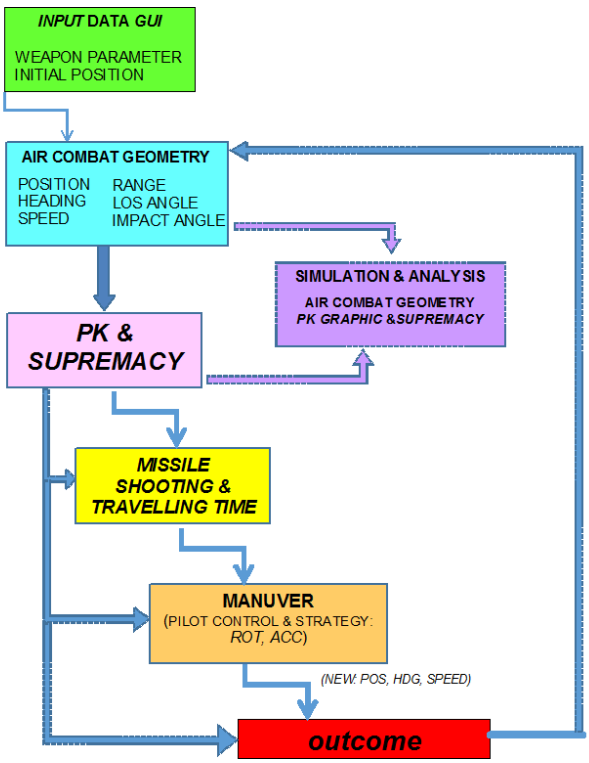


Figure 3. Simulation Logic

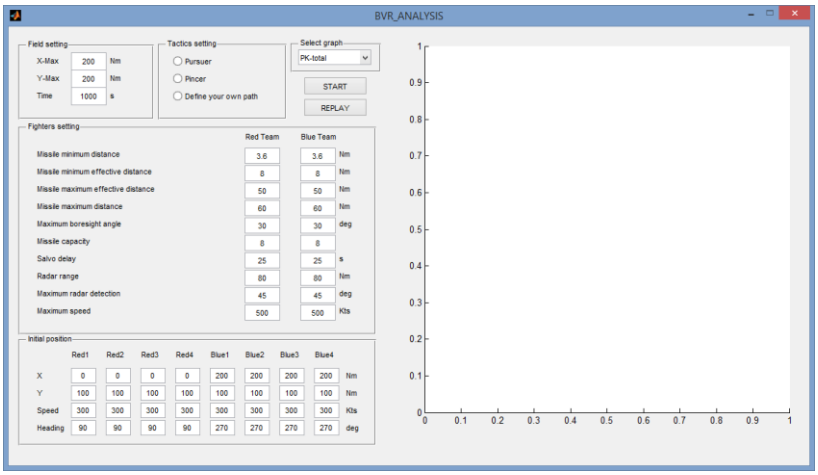


Figure 4. Simulation software main GUI

First, the simulation algorithm calculates the basic variables such as range, LOS angle, etc. These variables then are used to calculate the PK and Supremacy of the fighter with respect to the involved enemy aircrafts. When detecting multiple enemies/bandits, fighter will select a bandit, over which it has the highest supremacy, as the first target. Hence, when this bandit enter fighter's missile envelope, the PK is expected to increase significantly, providing a high possibility for a successful engagement.

In the simulation, fighter will shoot a missile when its PK reaches a minimum threshold value, which in this study is set to be 0.7. This value can be adjusted according to the user empirical knowledge or weapon specification. In addition to that, to avoid repeated shootings in a short time, an interval time between shoots can also be set for representing the cooldown time after each shot. Within that interval time fighter can execute a routine evading maneuver or keep chasing the enemy. Additionally the simulation programs will also calculate the prediction of time for the missile to reach its target.

Missile fly-out computation. Further, a fly-out models is used to simulate the movement of a missile along its way to reach the target. Probability of Kill is calculated based on the dynamic position change of the missile relative to the target at each time instant. This modeling technique updates the PK computation to adapt to bandit maneuver when avoiding the pursuing missile. Some problems associated with this modeling method are longer computing time and additional complexity in determining the relative position of the missile and the bandit. There 3 approaches which are developed in this work, each of them is illustrated in Figure 5.

In the first scheme, the PK only be calculated when the missile is fired. This method did not consider bandit maneuver executed after the time the missile is fired. As a result the PK is considered to remain constant regardless of bandit maneuver. The second scheme considers bandit maneuver, but the missile's fly-out is set to remain as it is when the missile fired. As a result the PK will be updated according to bandit's maneuver using the same missile envelope. The third scheme considers the bandit maneuver with respect to missile flyout which is progressively updated based on prediction of missile trajectory and missile available power at each time. In this scheme the PK values are computed based on predicted bandit-missile relative position and updated missile envelope.

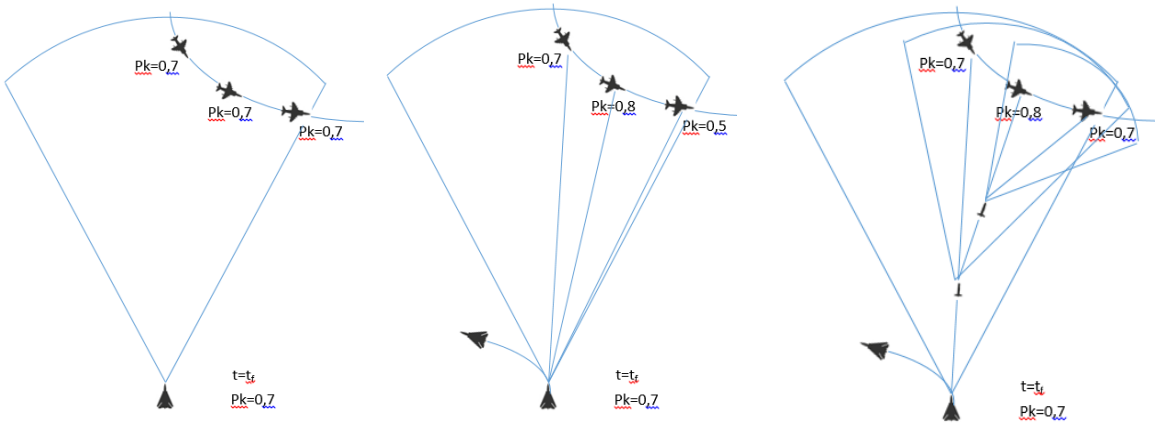
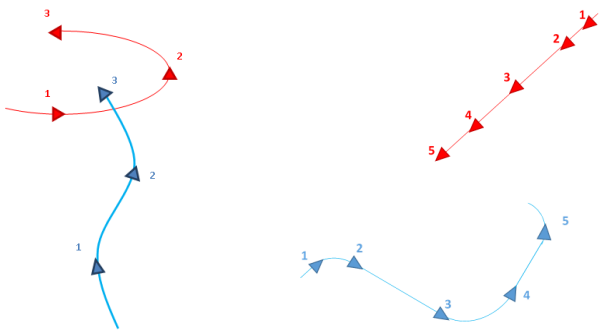


Figure 5. Missile Flyout Computation schemes (a) Fixed frame-initial time value only, (b) Fixed frame-iteratively updated value, (c) Updated frame-ieteratively updated value

When a missile reach a particular distance to the bandit, a method is required to determine whether the bandit is considered destroyed or not. A simple rule is employed in this work by setting the PK value as a threshold and generating a series of random number in the range of 0-1. Then the bandit is considered to be hit by the missile if the process produce a random number the value of which is less than or equal to the threshold value, other than that the missile is considered to miss the target. The simulation program also simulates a countermeasure actions, such as the deployment of flares or chaff, which will decrease the PK value (or increase the survivability of the bandit). In this work the simulation is set such that whenever a countermeasure is triggered then the PK will be halved. The probability or chance for the countermeasure itself to be activated is set to be 0.5.

Aircraft Maneuver. In the developed simulation scheme, fighter position and maneuver are determined by iteratively computing the position changes based on the updated information of relative position between fighter and bandit, and also the PK prediction. The maneuver is represented as the changes in speed variable and Rate of Turn (ROT) of all involved aircrafts (fighters and bandits). This paper only discusses two type of combat maneuvers associated with the discussed air combat modeling and simulation scheme. The first one is the Pursuer Tactics, which regulates the fighter maneuver for minimizing its LOS angle against the bandits. Thus, if the bandit is on the right side of the fighter, the fighter will generate positive ROT. Conversely, if the bandits are on the left side of the fighter, the fighter will generate negative ROT. In this mode the fighter will keep bandits to stay in front of it so that the PK-azimuth value is expected to be maximum.

The second maneuver is the Pincer Tactics which is based on the fact that a fighter who faces a bandit head-on has an impact angle supremacy lower than that of a fighter facing the bandit at a particular impact angle value.



(a) (b)

Figure 6. (a) Pursuer Maneuver, (b) Pincer Maneuver

Combat Simulation and Analysis

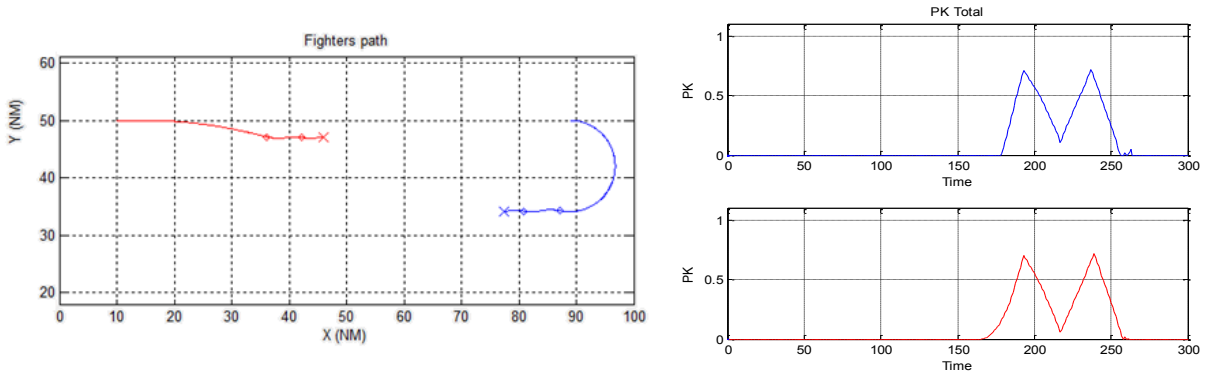
Using this software a set of combat situations are simulated and analyzed. Some scenarios are selected to be presented here.

Tail-chase Scenario. This Scenarios is used to determine how much the positional geometry influence the probability of kill (PK) value. In this 1 versus 1 case, both fighters (red & blue) involved in an engagement in a tail-chase position (blue fighter is at 0 ° azimuth relative to the red fighter with aspect angle 0°). At this simulation both aircrafts were assumed to have same weapon and radar systems, so the advantage will only be affected by the positional geometry of both fighters. In Figure 7a, the simulation shows that both fighter (red and blue) simultaneously detect each other due to the same radar detection capabilities. Fighter blue which is on the defensive position (low aspect angle relative to red fighter) then maneuvers to neutralize its position and retaliates. This maneuver is possible due to the fact that in BVR case, the fighter usually has enough turning room to neutralize the position. The PK graph (Figure 7b) also shows that the blue fighter is able to equalize its PK value to that of red fighter. Then, both fighters fire twice, each at an about the same time interval.

2v2 Combat. This Scenarios is a 2 versus 2 air combat on a head-on situation. In this simulation, the red team is set to have weapon with longer range capability. In return the blue team is equipped with better/longer radar detection range. This scenario is used to observe the influence of weapon and detection system parameters in a BVR air combat. The simulation shows that blue team reacts earlier due to its longer radar detection range. But longer weapon range allows red team to shoot first (see Figure 8a). Even though blue team then manages to destroy one of red team members, it is the red team who gain the final triumph. The change of PK values during the engagement can be seen in Figure 8b.

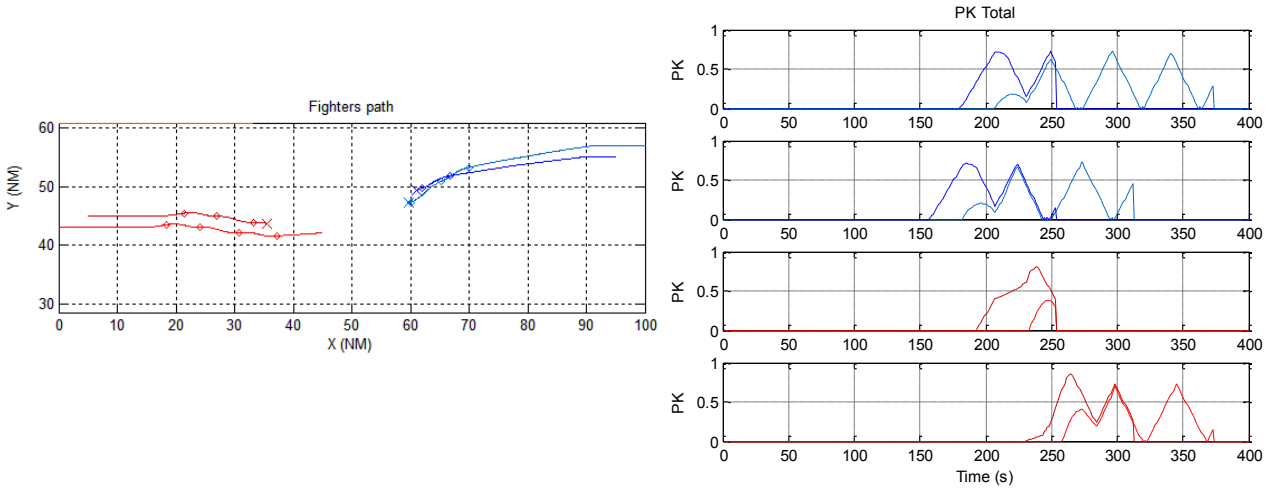
Conclusion

The simulation results show a good compatibility and comparison to some qualitative experience in the field, some of which can be related to empirical evaluation elaborated in [1]. The results also show that armament and radar/detection system play key role in controlling the combat.
Due to the limitations of existing information, a large part of results can only be assessed qualitatively, by observing the assessment of the expert (pilot), and cannot be assessed analytically yet. Formulation used in this work is in parametric form, appropriate corrections or adjustments may be required in accordance to known parameters and information about the involved systems (weapons, radar, etc).
In the case of multi aircrafts combat, strategy for distributing the targets to fighters, and rule for managing the shooting time will greatly determine the final results. Good coordination between fighters is substantial to select and implement the appropriate strategy.



(a). (b)

Figure 7. Tail-chase Scenario (a) Fighter-Bandit trajectories, (b) Fighter PK



(a) (b)

Figure 8. 2 versus 2 combat (a) Fighters-Bandits trajectories, (b) Fighters PK

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